

| Methods               | Hold-out     |              |              |              |              |              | Ensemble               |                        |                          |
|-----------------------|--------------|--------------|--------------|--------------|--------------|--------------|------------------------|------------------------|--------------------------|
|                       | -Res-50      | -Res-101     | -Res-152     | -IncRes-v2   | -Inc-v3      | -Inc-v4      | Inc-v3 <sub>ens3</sub> | Inc-v3 <sub>ens4</sub> | IncRes-v2 <sub>ens</sub> |
| I-FGSM                | 71.08        | 71.16        | 67.92        | 46.60        | 59.98        | 50.86        | 15.94                  | 8.54                   | 13.72                    |
| I-FGSM + <b>ours</b>  | 80.22        | 79.80        | 77.02        | 60.20        | 73.18        | 67.84        | 25.80                  | 13.56                  | 21.42                    |
| MI-FGSM               | 79.32        | 79.14        | 77.26        | 64.24        | 72.22        | 66.64        | 29.98                  | 16.66                  | 26.16                    |
| MI-FGSM + <b>ours</b> | <b>87.14</b> | <b>86.14</b> | <b>84.64</b> | <b>74.18</b> | <b>82.06</b> | <b>79.18</b> | <b>39.56</b>           | <b>21.24</b>           | <b>32.68</b>             |

Table 3: The attack rate (%) comparison of multi-model attack. “Ensemble” means attack all 6 naturally-trained models. “Hold-out” means attack 5 out of 6 models. The sign “-” indicates the name of the hold-out model.

| Methods                      | Black-box Attack |              |              |              | White-box Attack      |                          |              |              |
|------------------------------|------------------|--------------|--------------|--------------|-----------------------|--------------------------|--------------|--------------|
|                              | TsAIL            | iyswim       | Anil Thomas  | Average      | Inc-v3 <sub>adv</sub> | IncRes-v2 <sub>ens</sub> | Inc-v3       | Average      |
| No.1 Submission              | 13.60            | 43.20        | 43.90        | 33.57        | 94.40                 | 93.00                    | <b>97.30</b> | 94.90        |
| No.1 Submission+ <b>ours</b> | <b>14.80</b>     | <b>52.28</b> | <b>51.68</b> | <b>39.59</b> | <b>97.62</b>          | <b>96.00</b>             | 95.48        | <b>96.37</b> |

Table 4: The attack rate (%) comparison in the NeurIPS 2017 Adversarial Challenge.

MI-FGSM from 79.32 to 87.14. When testing on the three adversarially-trained networks, the improvement is more notable. These results further testify the ability of ghost networks to learn transferable adversarial examples.

### NeurIPS 2017 Adversarial Challenge

Finally, we evaluate our method in a benchmark test of the NeurIPS 2017 Adversarial Challenge (?). For performance evaluation, we use the top-3 defense submissions (black-box models), *i.e.*, TsAIL<sup>1</sup>, iyswim<sup>2</sup> and Anil Thomas<sup>3</sup>, and three official baselines (white-box models), *i.e.*, Inc-v3<sub>adv</sub>, IncRes-v2<sub>ens</sub> and Inc-v3. The test dataset contains 5000 images with the same 1000-class labels as ImageNet (?).

Following the experimental setting of the No.1 attack submission (?), we attack on an ensemble of Inc-v3, IncRes-v2, Inc-v4, Res-152, Inc-v3<sub>ens3</sub>, Inc-v3<sub>ens4</sub>, IncRes-v2<sub>ens</sub> and Inc-v3<sub>adv</sub> (?). The ensemble weights are set to 1/7.25 equally for the first seven networks and 0.25/7.25 for Inc-v3<sub>adv</sub>. The total iteration number is set to 10, and the maximum perturbation  $\epsilon$  is randomly selected from {4, 8, 12, 16}. The step size  $\alpha = \epsilon/10$ . The results are summarized in Table 4. Consistent with previous experiments, we observe that by applying ghost networks, the performance of the No. 1 submission can be significantly improved, especially with black-box attack. For example, the average performance of black-box attack is changed from 33.57 to 39.59, an improvement of 6.02. The most remarkable improvement is achieved when testing on iyswim, where ghost networks leads to an improvement of 9.08. This suggests that our proposed method can generalize well to other defense mechanisms.

### Conclusion

This paper focuses on learning transferable adversarial examples for adversarial attacks. We propose, for the first time, to exploit network erosion to generate a kind of virtual models called ghost networks. Ghost networks, together with the coupled longitudinal ensemble strategy, is an effective and efficient tool to improve existing methods in

learning transferable adversarial examples. Extensive experiments have firmly demonstrated the efficacy of ghost networks. Meanwhile, one can potentially apply erosion to residual unit by other methods or **densely** erode other typical layers (*e.g.*, batch norm (?) and relu (?)) **through** a neural network. We suppose these methods could improve the transferability as well, and leave these issues as future work.

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<sup>1</sup><https://github.com/lfz/Guided-Denoise>

<sup>2</sup>[https://github.com/cihangxie/NIPS2017\\_adv\\_challenge\\_defense](https://github.com/cihangxie/NIPS2017_adv_challenge_defense)

<sup>3</sup><https://github.com/anlthms/nips-2017/tree/master/mmd>

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